

BHB

CIVIL ENGINEERING REVIEW

SPECIAL PROFILE

YOUSEF BAHRAMBEIGI

Engineering structures
through optimization, AI and
open tools



PhD Student - Civil Engineering, Structures

Researcher | Lecturer | Consultant Engineer



ISSUE 01 / 2026

Where structural mechanics meets computational intelligence

Yousef Bahram Beigi is a civil and structural engineering researcher, lecturer and consultant whose work connects finite-element analysis, metaheuristic optimization, engineering education and practice-oriented digital systems.

His portfolio moves across three scales: the member-level logic of truss sizing, the system-level behavior of space structures, and the human-scale challenge of making computational engineering more accessible, reproducible and useful.



CURRENT PROFESSIONAL IDENTITY

PhD Student in Civil Engineering -
Structures
Researcher | Lecturer | Consultant
Engineer
Main Inspector, West Azerbaijan
Construction Engineering Organization

**DESIGN.
ANALYZE.
OPTIMIZE.
INNOVATE.**

Engineering as a transparent, evidence-led practice.

A portfolio built through analysis, research and public service



Lighter structures. Stronger evidence. Wider access.

0

STRUCTURES

1

Optimization under constraints

Minimum-weight steel design is treated as a constrained search problem governed by stress, displacement, stability and discrete section choices.

0

COMPUTATION

2

FEM and nonlinear behavior

Finite-element models connect design variables to structural response, while nonlinear analysis reveals mechanisms hidden by linear assumptions.

0

IMPACT

3

Open engineering education

Transparent code, benchmark problems and interactive learning tools reduce the distance between theory and reproducible engineering practice.



A transparent pipeline for multi-objective steel-truss optimization

The doctoral program focuses on improving metaheuristic search for discrete sizing of steel trusses while keeping every result traceable to structural constraints and benchmark evidence.

PROPOSED WORKFLOW



PRIMARY OBJECTIVES

Minimize structural weight

Reduce seismic-cost proxies

Quantify reliability and search consistency

The contribution is not a single best number. It is a reproducible decision trail from design variables to structural performance.

Benchmarking is the discipline behind the algorithm

A six-algorithm MATLAB framework compares search behavior on a classic 10-bar truss under engineering constraints.

ALGORITHMS

**GA / PSO / DE
SA / HS / TLBO**

ENGINEERING FORMULATION

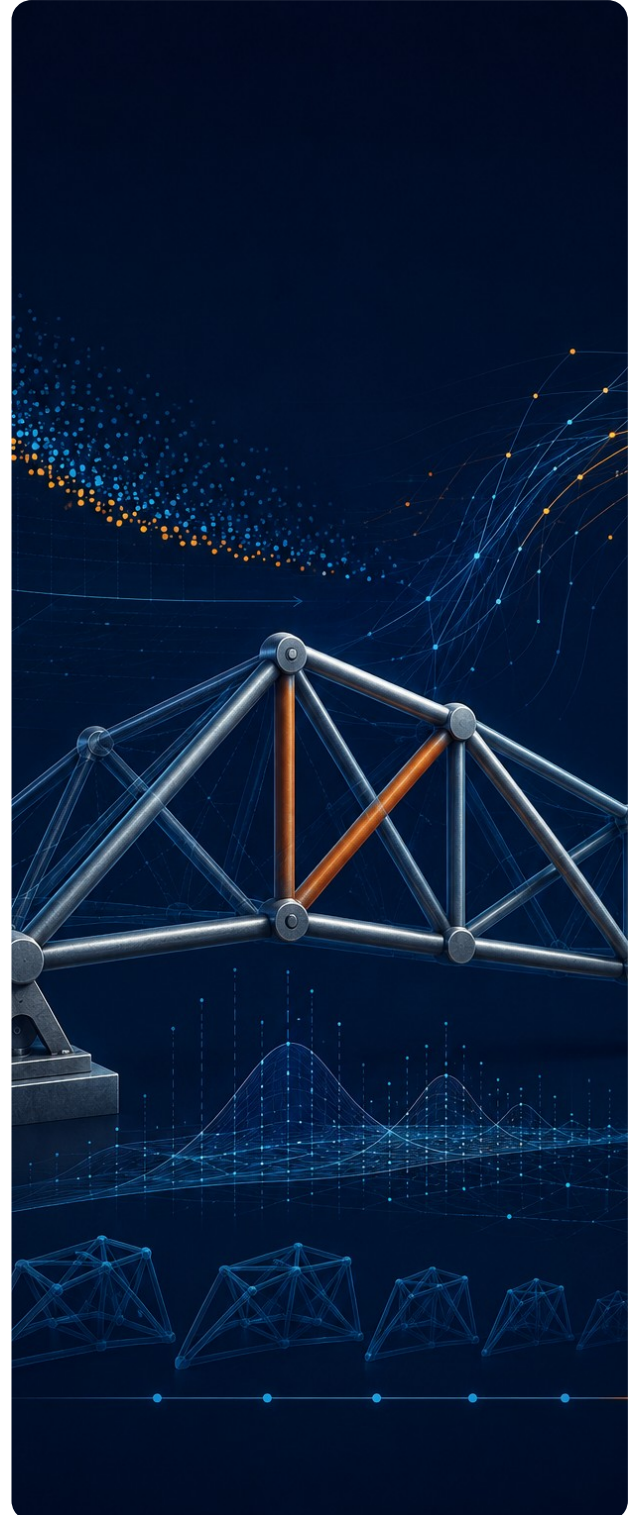
Objective
Minimum structural weight

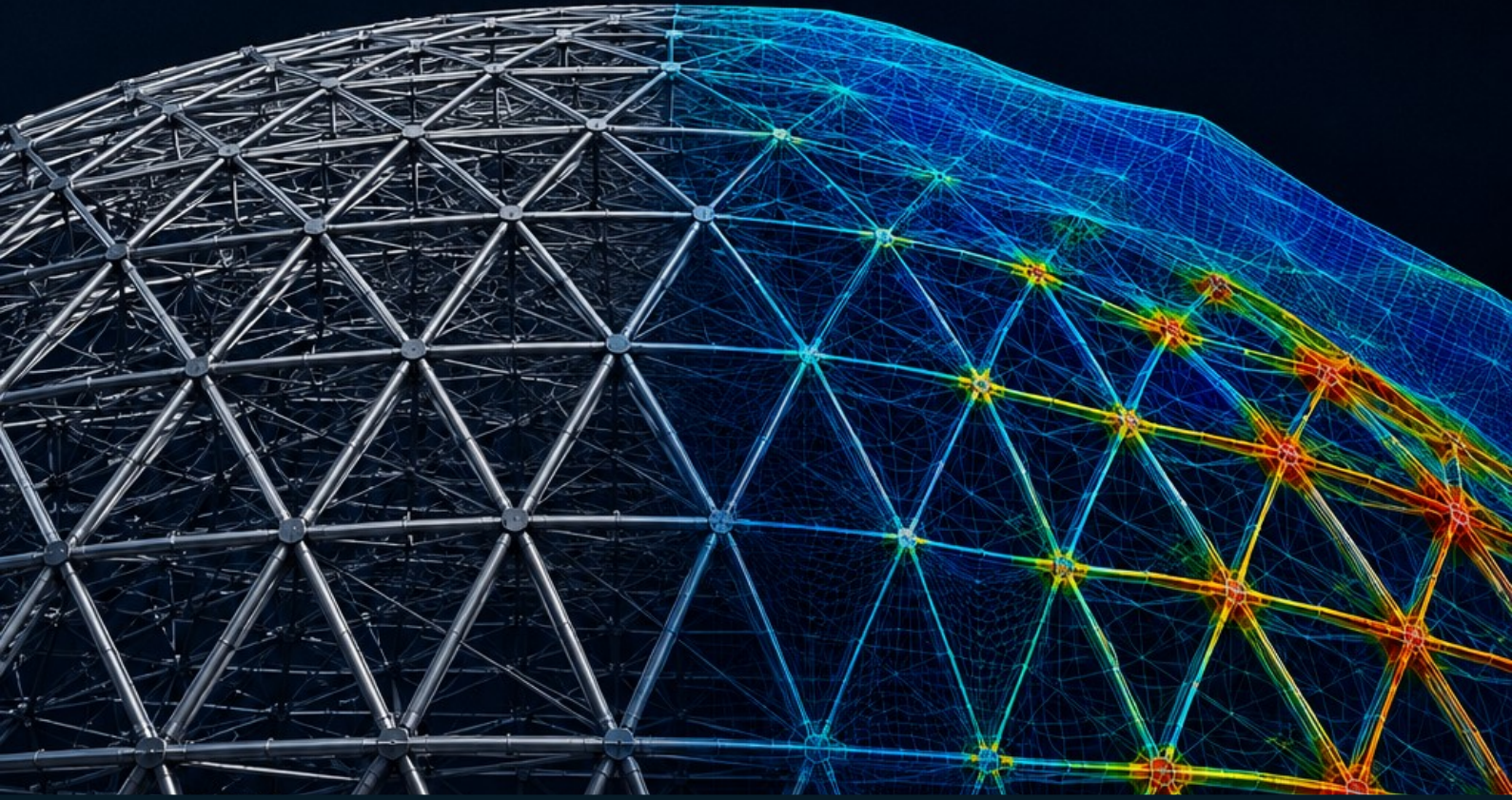
Constraints
Member stress and nodal displacement

Analysis
Direct stiffness method

Evidence
Convergence, runtime, feasibility and ranking

Repository: github.com/y0bahrambeigi/bhb





NONLINEAR SPACE STRUCTURES

Geometry becomes performance in the double-layer dome

The 2023 study modeled double-layer braced domes in ANSYS, compared linear and nonlinear response, and examined how span, rise-to-span and thickness-to-span choices influence deflection and design behavior.

20 m
30 m
50 m
80 m

MODELED SPANS

Papers, software and persistent identifiers form one research trail

2026

Python Learning Studio Ultra

Archived software, Version 1.0.1
DOI 10.5281/zenodo.21960225

Python Learning Studio Ultra: An Interactive Cross-Platform Environment for Python Programming, Scientific Computing, Artificial Intelligence, and Computational Civil Engineering

Yousef Bahram Beigi

PhD Student in Structural Engineering | Computational Engineering
Software DOI: 10.5281/zenodo.21960225 | Technical Paper - Version 1.0 | 2026

Abstract

Python Learning Studio Ultra is an interactive, cross-platform learning environment designed to integrate Python programming education with scientific computing, artificial intelligence concepts, and computational civil engineering. The platform combines an editor-oriented learning workflow, browser-based Python execution, numerical and plotting support, interactive exercises, staged quizzes, progress mechanisms, and engineering-oriented examples. This technical paper documents the educational rationale, system architecture, principal functional modules, reproducibility strategy, and intended use in engineering education. Particular emphasis is placed on lowering the transition cost from introductory programming to computational problem solving in civil and structural engineering. The software release is preserved through Zenodo and is identified by DOI 10.5281/zenodo.21960225.

Keywords: Python; engineering education; scientific computing; civil engineering; structural engineering; artificial intelligence; interactive learning; PKA; computational engineering

Python Learning Studio Ultra | Yousef Bahram Beigi | DOI: 10.5281/zenodo.21960225 | Page 1

2026

Technical report

Interactive Python, scientific computing, AI and computational civil engineering
DOI 10.5281/zenodo.21971320

2023

Non-linear analysis of double-layer dome space structures

ANSYS-based comparative study with 20-80 m spans

2022

Structural optimization of truss members

Metaheuristic-algorithm research record listed in the public academic portfolio

Software is part of the scholarly record when versions, source and citation metadata remain verifiable.

Python Learning Studio Ultra

A cross-platform environment designed to connect introductory code with scientific computing and structural-engineering applications.

01 Monaco-style code editor

02 Browser Python with Pyodide

03 NumPy and Matplotlib workflows

04 Exercises, tests, quizzes and progress

05 Civil Engineering Lab examples

06 PWA, desktop and mobile pathways

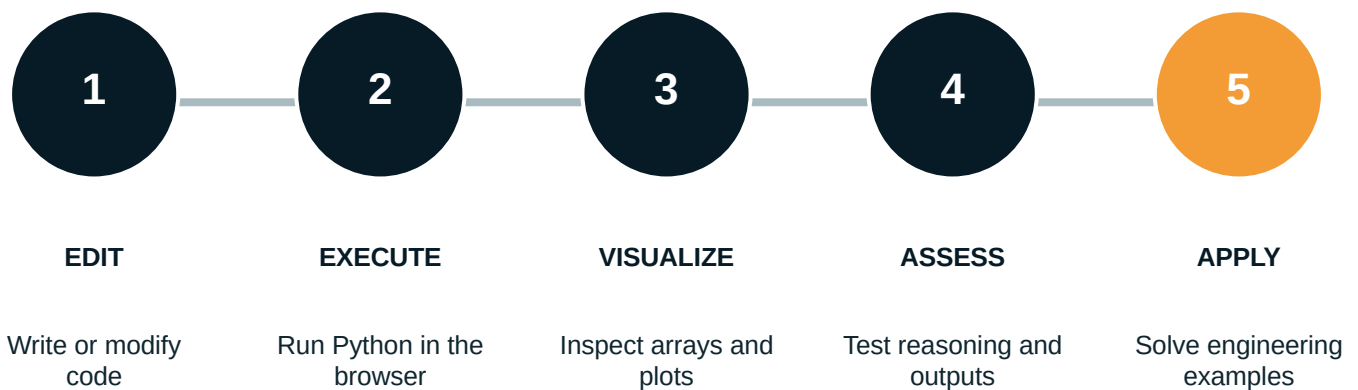
[DOI 10.5281/zenodo.21960225](https://doi.org/10.5281/zenodo.21960225)

y0bahrambeigi.github.io/bhb/python-learning-studio-pro

The platform turns code execution into an engineering learning cycle

The educational design keeps explanation, execution, feedback and application in one continuous environment.

LEARNING CYCLE



THE RESEARCH QUESTION

Does an integrated environment improve the transition from introductory programming to reproducible computational engineering?

Future evaluation should measure learning outcomes, compatibility, accessibility and the correctness of benchmark solutions - not feature count alone.

BHB Smart Structure Lab converts expertise into an operating system

BHB Business OS 3.0 organizes the non-technical work that allows engineering expertise to scale: clients, finances, content, growth plans and offline continuity.

0 PIPELINE

1 Engineering CRM

Track client opportunities, stages, values and next actions in one practical workspace.

0 CONTROL

2 Financial notebook

Record revenue and cost, calculate net performance and export local JSON backups.

0 VISIBILITY

3 Content studio

Plan publishing, apply reusable AI prompts and create a repeatable engineering content workflow.

0 RESILIENCE

4 Private offline PWA

Store working data on the device, support installation and maintain continuity across common platforms.

bhb-smart-structure-lab.yousefbhb.chatgpt.site

Mohandesyar AI strengthens the evidence trail

A field documentation and reporting assistant for civil and architectural workflows.

- Project dossier and inspection stages
- Local photo and video evidence records
- Time and location traceability
- Formal printable reporting
- Compliance warnings before sign-off
- Local-first browser storage

DOI [10.5281/zenodo.22035833](https://doi.org/10.5281/zenodo.22035833)

Professional judgment remains the final control point.



BHB Engineering Token tests transparent coordination on Sepolia



TESTNET DESIGN

100M

initial BHB supply

CAPPED ISSUANCE

Up to 20 million additional tokens, with a hard lifetime ceiling of 120 million.

CONTROL MODEL

Emergency pause
Two-step ownership transfer
Planned Safe multisig governance
Public source and test coverage

A testnet case study in bounded supply, emergency controls and governance transition.

PROJECT STATUS

Prototype environment • verification before promotion

SEPOLIA TESTNET

Technology preview - not an investment offer

Technical practice carries responsibility to institutions and the public



یوسف بهرام بیگی

دکتری تخصصی مهندسی عمران - سازه
پژوهشگر | مدرس | مهندس مشاور
بازرس اصلی کل سازمان نظام مهندسی ساختمان استان آذربایجان غربی
داور حقوقی استان آذربایجان غربی - ارومیه - کد: ۱۶۴۷۳۴
مدرس بین المللی ITIC با کد: ۱۸۶۹۳
ASCE(American Society of Civil Engineers)



YUSEF BAHRAM BEIGI

PhD in Civil Engineering – Structures
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بهبودسازی سازه‌ها



تحلیل و طراحی سازه



الگوریتم‌های فراابتکاری و هوش مصنوعی



مدل‌سازی و تحلیل عددی (FEM)

EXPERTISE

-  Structural Optimization
-  Metaheuristic Algorithms
-  Artificial Intelligence in Engineering
-  Finite Element Analysis (FEM)
-  Seismic Performance & Resilience
-  Research & Academic Training

Y. Bahram Beigi



DESIGN • ANALYZE • OPTIMIZE
INNOVATE • IMPACT

«دانش، دقت و نوآوری برای ساختن آینده‌ای پایدار»

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SCAN ME

MAIN INSPECTOR

West Azerbaijan Construction Engineering Organization

The role brings financial oversight, governance, compliance and accountability into the same professional identity as research and design.

PROFESSIONAL NETWORK

ASCE member
International instructor
credential - ITIC 18693
Researcher, lecturer and
consultant engineer

**Engineering credibility is built
through competence, transparency
and documented judgment.**

Open work. Verifiable records. Human judgment.

The next phase is deeper validation: independent software testing, reproducible structural benchmarks, educational pilots, digital-twin research and stronger links between field evidence and finite-element models.

SELECTED PUBLIC LINKS

github.com/y0bahrambeigi
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researchgate.net/profile/Yousef_Beigi

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YOUSEF BAHRAM BEIGI

Civil & Structural Engineering | Research | AI | Optimization